Topic: The matrix exponential

Question: Use the matrix exponential to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} \cosh t \\ \sinh t \end{bmatrix}$$

Answer choices:

A $\vec{x} = \begin{bmatrix} c_1 + t \\ c_2 \end{bmatrix} \sinh t + \begin{bmatrix} c_2 \\ c_1 + t \end{bmatrix} \cosh t$

B $\vec{x} = \begin{bmatrix} c_2 \\ c_1 + t \end{bmatrix} \sinh t + \begin{bmatrix} c_1 + t \\ c_2 \end{bmatrix} \cosh t$

C $\vec{x} = \begin{bmatrix} c_1 t \\ c_2 \end{bmatrix} \sinh t + \begin{bmatrix} c_2 \\ c_1 t \end{bmatrix} \cosh t$

D $\vec{x} = \begin{bmatrix} c_2 \\ c_1 t \end{bmatrix} \sinh t + \begin{bmatrix} c_1 t \\ c_2 \end{bmatrix} \cosh t$

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Solution: B

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s-0 & 0-1 \\ 0-1 & s-0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & -1 \\ -1 & s \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{cc|cc} s & -1 & 1 & 0 \\ -1 & s & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} -1 & s & 0 & 1 \\ s & -1 & 1 & 0 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & -s & 0 & -1 \\ s & -1 & 1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & -s & 0 & -1 \\ 0 & s^2-1 & 1 & s \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & -s & 0 & -1 \\ 0 & 1 & \frac{1}{s^2-1} & \frac{s}{s^2-1} \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{s}{s^2-1} & \frac{1}{s^2-1} \\ 0 & 1 & \frac{1}{s^2-1} & \frac{s}{s^2-1} \end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

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$$(sI - A)^{-1} = \begin{bmatrix} \frac{s}{s^2-1} & \frac{1}{s^2-1} \\ \frac{1}{s^2-1} & \frac{s}{s^2-1} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI - A)^{-1}) = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix}$$

and then say that this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the complementary solution will be

$$\overrightarrow{x_c} = e^{At}C$$

$$\overrightarrow{x_c} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\overrightarrow{x_c} = \begin{bmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{bmatrix}$$

To find the particular solution, we'll need e^{-As}, which we find by making the substitution $t = s$ into e^{At},

$$e^{As} = \begin{bmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{bmatrix}$$

and then calculating the inverse of this resulting e^{As}.

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$$\left[\begin{array}{cc|cc}\cosh s & \sinh s & 1 & 0 \\ \sinh s & \cosh s & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & \frac{\sinh s}{\cosh s} & \frac{1}{\cosh s} & 0 \\ \sinh s & \cosh s & 0 & 1\end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc}1 & \frac{\sinh s}{\cosh s} & \frac{1}{\cosh s} & 0 \\ 0 & \frac{1}{\cosh s} & -\frac{\sinh s}{\cosh s} & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & \frac{\sinh s}{\cosh s} & \frac{1}{\cosh s} & 0 \\ 0 & 1 & -\sinh s & \cosh s\end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc}1 & 0 & \cosh s & -\sinh s \\ 0 & 1 & -\sinh s & \cosh s\end{array}\right]$$

So e^{-As} is given by this resulting matrix.

$$e^{-As} = \begin{bmatrix}\cosh s & -\sinh s \\ -\sinh s & \cosh s\end{bmatrix}$$

We'll find $F(s)$ by substituting $t = s$ into F.

$$F(s) = \begin{bmatrix}\cosh s \\ \sinh s\end{bmatrix}$$

Therefore, the particular solution will be

$$\overrightarrow{x_p} = e^{At}\int_{t_0}^{t} e^{-As}F(s)\,ds$$

$$\overrightarrow{x_p} = \begin{bmatrix}\cosh t & \sinh t \\ \sinh t & \cosh t\end{bmatrix}\int_0^t \begin{bmatrix}\cosh s & -\sinh s \\ -\sinh s & \cosh s\end{bmatrix}\begin{bmatrix}\cosh s \\ \sinh s\end{bmatrix}\,ds$$

$$\overrightarrow{x_p} = \begin{bmatrix}\cosh t & \sinh t \\ \sinh t & \cosh t\end{bmatrix}\int_0^t \begin{bmatrix}\cosh^2 s - \sinh^2 s \\ -\sinh s\cosh s + \sinh s\cosh s\end{bmatrix}\,ds$$

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$$\overrightarrow{x_p} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix} \int_0^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} \, ds$$

Now we'll integrate and then evaluate on $[0,t]$.

$$\overrightarrow{x_p} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix} \begin{bmatrix} s \\ 0 \end{bmatrix} \Bigg|_0^t$$

$$\overrightarrow{x_p} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix} \left(\begin{bmatrix} t \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right)$$

$$\overrightarrow{x_p} = \begin{bmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{bmatrix} \begin{bmatrix} t \\ 0 \end{bmatrix}$$

Finally, we can do the last matrix multiplication to get $\overrightarrow{x_p}$.

$$\overrightarrow{x_p} = \begin{bmatrix} \cosh t(t) + \sinh t(0) \\ \sinh t(t) + \cosh t(0) \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} t \cosh t \\ t \sinh t \end{bmatrix}$$

Then the general solution is the sum of the complementary and particular solutions.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = \begin{bmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{bmatrix} + \begin{bmatrix} t \cosh t \\ t \sinh t \end{bmatrix}$$

$$\overrightarrow{x} = \begin{bmatrix} c_2 \\ c_1 + t \end{bmatrix} \sinh t + \begin{bmatrix} c_1 + t \\ c_2 \end{bmatrix} \cosh t$$

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Topic: The matrix exponential

Question: Use the matrix exponential to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = \begin{bmatrix} c_1 \\ -2c_1t + c_2 \\ 3c_1t^2 + (5c_1 + c_2)t + c_3 \end{bmatrix}$$

B $$\vec{x} = \begin{bmatrix} c_1 \\ -2c_1t + c_2 \\ 3c_1t^2 + 5c_2t + c_3 \end{bmatrix}$$

C $$\vec{x} = \begin{bmatrix} c_1 \\ 3c_1t + c_2 \\ \frac{3}{2}c_1t^2 + (5c_1 + c_2)t + c_3 \end{bmatrix}$$

D $$\vec{x} = \begin{bmatrix} c_1 \\ 3c_1t + c_2 \\ \frac{3}{2}c_1t^2 + 5c_2t + c_3 \end{bmatrix}$$

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Solution: C

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s-0 & 0-0 & 0-0 \\ 0-3 & s-0 & 0-0 \\ 0-5 & 0-1 & s-0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ -3 & s & 0 \\ -5 & -1 & s \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{ccc|ccc} s & 0 & 0 & 1 & 0 & 0 \\ -3 & s & 0 & 0 & 1 & 0 \\ -5 & -1 & s & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ -3 & s & 0 & 0 & 1 & 0 \\ -5 & -1 & s & 0 & 0 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ 0 & s & 0 & \frac{3}{s} & 1 & 0 \\ -5 & -1 & s & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ 0 & s & 0 & \frac{3}{s} & 1 & 0 \\ 0 & -1 & s & \frac{5}{s} & 0 & 1 \end{array}\right]$$

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$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{s^2} & \frac{1}{s} & 0 \\ 0 & -1 & s & \frac{5}{s} & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{s^2} & \frac{1}{s} & 0 \\ 0 & 0 & s & \frac{5s+3}{s^2} & \frac{1}{s} & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{s^2} & \frac{1}{s} & 0 \\ 0 & 0 & 1 & \frac{5s+3}{s^3} & \frac{1}{s^2} & \frac{1}{s} \end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s} & 0 & 0 \\ \frac{3}{s^2} & \frac{1}{s} & 0 \\ \frac{5s+3}{s^3} & \frac{1}{s^2} & \frac{1}{s} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI - A)^{-1}) = \begin{bmatrix} 1 & 0 & 0 \\ 3t & 1 & 0 \\ \frac{3}{2}t^2 + 5t & t & 1 \end{bmatrix}$$

and then say that this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} 1 & 0 & 0 \\ 3t & 1 & 0 \\ \frac{3}{2}t^2 + 5t & t & 1 \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the general solution to the homogeneous system is

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$$\vec{x} = e^{At}C$$

$$\vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ 3t & 1 & 0 \\ \frac{3}{2}t^2 + 5t & t & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} c_1 \\ 3c_1t + c_2 \\ \frac{3}{2}c_1t^2 + (5c_1 + c_2)t + c_3 \end{bmatrix}$$

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Topic: The matrix exponential

Question: Use an inverse Laplace transform to calculate the matrix exponential.

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix}$$

Answer choices:

A $$\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix} \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) - 3\sinh\left(\frac{\sqrt{5}}{2}t\right) & 2\sinh\left(\frac{\sqrt{5}}{2}t\right) \\ -2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) + 3\sinh\left(\frac{\sqrt{5}}{2}t\right) \end{bmatrix}$$

B $$\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix} \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) + 3\sinh\left(\frac{\sqrt{5}}{2}t\right) & -2\sinh\left(\frac{\sqrt{5}}{2}t\right) \\ 2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) - 3\sinh\left(\frac{\sqrt{5}}{2}t\right) \end{bmatrix}$$

C $$\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix} \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) & 2\sinh\left(\frac{\sqrt{5}}{2}t\right) \\ -2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) \end{bmatrix}$$

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D $$\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix}\sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right) & -2\sinh\left(\frac{\sqrt{5}}{2}t\right)\\ 2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right)\end{bmatrix}$$

Solution: B

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} - \begin{bmatrix}3 & -1\\ 1 & 0\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & 0\\ 0 & s\end{bmatrix} - \begin{bmatrix}3 & -1\\ 1 & 0\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s-3 & 0-(-1)\\ 0-1 & s-0\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s-3 & 1\\ -1 & s\end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{cc|cc}s-3 & 1 & 1 & 0\\ -1 & s & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & \frac{1}{s-3} & \frac{1}{s-3} & 0\\ -1 & s & 0 & 1\end{array}\right]$$

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$$\rightarrow \begin{bmatrix} 1 & \frac{1}{s-3} & | & \frac{1}{s-3} & 0 \\ 0 & \frac{s(s-3)+1}{s-3} & | & \frac{1}{s-3} & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{s-3} & | & \frac{1}{s-3} & 0 \\ 0 & 1 & | & \frac{1}{s(s-3)+1} & \frac{s-3}{s(s-3)+1} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & | & \frac{s}{s^2-3s+1} & -\frac{1}{s^2-3s+1} \\ 0 & 1 & | & \frac{1}{s^2-3s+1} & \frac{s-3}{s^2-3s+1} \end{bmatrix}$$

Before we can apply an inverse transform to the inverse matrix, we need to rewrite the entries to prepare them for the inverse Laplace transform.

$$(sI-A)^{-1} =$$

$$\begin{bmatrix} \frac{s-\frac{3}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}+\frac{3\sqrt{5}}{5}\left(\frac{\frac{\sqrt{5}}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}\right) & -\frac{2\sqrt{5}}{5}\left(\frac{\frac{\sqrt{5}}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}\right) \\ \frac{2\sqrt{5}}{5}\left(\frac{\frac{\sqrt{5}}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}\right) & \frac{s-\frac{3}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}-\frac{3\sqrt{5}}{5}\left(\frac{\frac{\sqrt{5}}{2}}{\left(s-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}\right) \end{bmatrix}$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$\mathscr{L}^{-1}((sI-A)^{-1}) =$$

$$\begin{bmatrix} e^{\frac{3}{2}t}\cosh\left(\frac{\sqrt{5}}{2}t\right)+\frac{3\sqrt{5}}{5}e^{\frac{3}{2}t}\sinh\left(\frac{\sqrt{5}}{2}t\right) & -\frac{2\sqrt{5}}{5}e^{\frac{3}{2}t}\sinh\left(\frac{\sqrt{5}}{2}t\right) \\ \frac{2\sqrt{5}}{5}e^{\frac{3}{2}t}\sinh\left(\frac{\sqrt{5}}{2}t\right) & e^{\frac{3}{2}t}\cosh\left(\frac{\sqrt{5}}{2}t\right)-\frac{3\sqrt{5}}{5}e^{\frac{3}{2}t}\sinh\left(\frac{\sqrt{5}}{2}t\right) \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI-A)^{-1}) =$$

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$$\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix}\sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right)+3\sinh\left(\frac{\sqrt{5}}{2}t\right) & -2\sinh\left(\frac{\sqrt{5}}{2}t\right)\\ 2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right)-3\sinh\left(\frac{\sqrt{5}}{2}t\right)\end{bmatrix}$$

and then say that this result is the matrix exponential.

$$e^{At}=\frac{\sqrt{5}}{5}e^{\frac{3}{2}t}\begin{bmatrix}\sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right)+3\sinh\left(\frac{\sqrt{5}}{2}t\right) & -2\sinh\left(\frac{\sqrt{5}}{2}t\right)\\ 2\sinh\left(\frac{\sqrt{5}}{2}t\right) & \sqrt{5}\cosh\left(\frac{\sqrt{5}}{2}t\right)-3\sinh\left(\frac{\sqrt{5}}{2}t\right)\end{bmatrix}$$

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